

Task/Exercise	Why?	What?	How? (Pointers)	When?	Expected Results	Reference Links
Cloud Concepts & Free Tiers	Gain cloud context	Create trial cloud account	AWS/GCP/Azure free tier, deploy sample VM	Week 1	Account setup confirmation	AWS Free Tier - https://aws.amazon.com/free/ , Azure Learn - https://learn.microsoft.com/en-us/training/
Linux Basics & Shell Scripting	Foundational Linux skill	Linux install, bash basics	Try commands, write simple scripts	Week 2	Script files	Codecademy Linux - https://www.codecademy.com/learn/learn-the-command-line
VMs Introduction	Understand classic Windows/Linux VMs	Launch/manage VM	Deploy, terminate VM, SSH	Week 3	VM access logs	Azure VM Docs - https://learn.microsoft.com/en-us/azure/virtual-machines/
Docker & Container Fundamentals	Containers essentials	Build/run docker container	Dockerfile, lifecycle commands	Week 4	Docker image and logs	Docker 101 - https://www.docker.com/101-tutorial
Kubernetes Basics	Industry standard orchestration	Create cluster, deploy app	Minikube install, kubectl commands	Week 5	Cluster logs, deployed app	Kubernetes Tutorials - https://kubernetes.io/docs/tutorials/
Helm Charts & Cloud Ecosystem	Simplify app deployment	Install/apply Helm charts	Helm repo, values.yaml customization	Week 6	Chart config and release notes	Helm Docs - https://helm.sh/docs/intro/using_helm/
Cloud Storage, DB, IAM, Networking	Core platform skills	S3/GCS, RDS, IAM usage	Storage buckets, DB launch, IAM policies	Week 7	Infrastructure provisioning	AWS Docs - https://docs.aws.amazon.com/
IaC with Terraform	Automation & repeatability	Provision infra with code	Terraform scripts, terraform apply	Week 8-9	Terraform state files, output	Terraform Getting Started - https://developer.hashicorp.com/terraform/tutorials/aws-get-started
Monitoring and Logging	Ops knowledge	Setup monitoring & alerting	CloudWatch, Grafana dashboards	Week 10	Alert logs and metrics	AWS CloudWatch - https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/WhatIsCloudWatch.html

Area	Certification	Link	Resources
Cloud	AWS Certified Cloud Practitioner	https://aws.amazon.com/certification/certified-cloud-practitioner/	AWS Training and Certification portal, AWS Cloud tutorials and labs, AWS free tier for hands-on practice
Cloud	Microsoft Certified: Azure Fundamentals	https://learn.microsoft.com/en-us/certifications/exams/az-900/	Microsoft Learn platform, Azure documentation, free learning paths
Cloud	Google Associate Cloud Engineer	https://cloud.google.com/certification/cloud-engineer	Google Cloud Certification, Coursera courses, Google Cloud docs